Fiscal Year:	FY 2019	**Task Last Updated:**	FY 10/17/2018
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Project Title:	Dynamic Self-assembly Driven by Time with Varying Fields		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
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Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	02139-4301	**Congressional District:**	7
Comments:			
Project Type:	Ground,Physical Sciences Informatics (PSI)	**Solicitation / Funding Source:**	2015-16 Physical Sciences NNH15ZTT001N-15PSI-C: Use of the NASA Physical Sciences Informatics System – Appendix C
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No. of PhD Candidates:	1	**No. of Master' Degrees:**	
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No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:	McQuillen, John	**Contact Phone:**	216-433-2876
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	October 2018 report: The project began this year and a 4th year graduate student in chemical engineering at MIT, Zachary Sherman, started work on the proposed tasks.		
COI Name (Institution):			
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Task Description:	This proposal is a research plan for using theory and simulation to analyze and extend the InSPACE 1, 2, 3, and 3+ investigations of the self-assembly of magnetically responsive nanoparticles in a time-varying magnetic field. The original work found that self-assembly was greatly accelerated by utilizing a simple open-loop control scheme of the magnetic interactions. This methodology offers one solution to the difficult engineering challenge of developing scalable and cost-effective fabrication methods for structured nanomaterials with proven applications in optoelectronics, energy storage and harvesting, biological sensing, 3D printing, and high-performance coatings. However, because of limited spatial and temporal resolution in the experiments, detailed analysis of the self-assembled structures was impossible. A deeper understanding of out-of-equilibrium self-assembly demonstrated in the InSPACE suite is needed before it can be leveraged to design processes that rapidly produce high quality nanomaterials. The proposed work has two main objectives. First, we will take advantage of insights developed since the InSPACE experiments for other out-of-equilibrium assemblies to re-evaluate the InSPACE data in the Physical Sciences Informatics (PSI) system. This will reveal new results that the original InSPACE team was unable to obtain. Second, we will use the InSPACE data in the PSI system to develop and validate a computational model of nanoparticle self-assembly in time-varying fields and use that model to extend the investigation of toggled magnetic field assembly beyond the parameter space explored in the experimental suite. This will aid in developing a predictive theoretical framework for dynamic self-assembly that can facilitate design of rapid nanomaterial syntheses. The work will utilize GPU-accelerated discrete element simulations of particle self-assembly driven by magnetic fields with detailed models of the magnetic and hydrodynamic interactions among the particles. The Principal Investigator's (PI) laboratory has recently developed new algorithms for modeling these interactions that have accelerated simulations by roughly a factor of 100 times. Consequently, discrete element simulations of the hierarchical self-assemblies observed in the InSPACE experiments are possible. The video micrographs from InSPACE are of limited spatial and temporal resolution; therefore, simulations capable of resolving the particle structure and dynamics on a local scale while replicating structure and growth kinetics on larger length scales and longer time scales provide additional insight into the InSPACE data. The ability to translate InSPACE microgravity data for the many-bodied, dynamic interactions among self-assembling particles into well-validated models for the same experiments will enable simulations to predict and then design dynamic self-assembly processes terrestrially as well as during continued space operations. The PI's laboratory has already demonstrated a predictive framework for other types of toggled self-assembly and is currently developing theories for the InSPACE assembly that can be used to predict the outcome and quantify the robustness of dynamical self-assembly processes. Incorporation of microgravity data into this theory is essential for testing its predictive capability. The PI was part of the original InSPACE 2 team as a post-doctoral scholar at the University of Delaware before starting as an assistant professor at Massachusetts Institute of Technology (MIT). He is first author of the two published papers on InSPACE experiments and is intimately familiar with the project. He has also published several papers extending toggled self-assembly to ground experiments and to simulations of new systems. One MIT graduate student, Zachary Sherman, currently completing a Ph.D. focused on toggled self-assembly processes will assist in the research. Their extensive experience makes the investigators uniquely qualified to re-examine and extend the InSPACE analysis.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The proposed work focuses on far from equilibrium methods for assembling nano-particles. The goal is to discover routes to self assembly free of kinetic constraints introduced by equilibrium thermodynamics. If successful, these pathways will enable the production of microstructures materials with few defects at rates not accessible by current technological platforms. The potential implications for these materials technologies span advance manufacturing in the form of 3D printing, the development of high performance coatings, and the creation of functional or tunable photonics and phononic materials for micro-devices.
	A diverse set of functional materials can be fabricated by assembling dispersions of colloids and nanoparticles. For polarizeable, paramagnetic colloids, the interactions among particles can be controlled using an external magnetic field, so the field can be used to facilitate self-assembly processes. Magnetic field-directed self-assembly is particularly effective because the external field is easy to control and time scales for magnetic relaxation inside particles are small compared to time scales for particle motion, so dispersions respond essentially instantaneously to the field. The assembled structures respond to changes in the field, allowing their properties to be controlled on-the-fly. Accurate predictions for the equilibrium phase diagram are vital to facilitate rational design and use of paramagnetic nanomaterials, but the phase behavior is difficult to calculate theoretically because of the many-bodied, long-ranged nature of the induced magnetic interactions. Though there are predictions based on the approximation that all particles possess the same dipole moment, regardless of their local organization or the bulk density, these predictions do not agree with the assembly behavior observed in experiments. We have developed a new theoretical and computational thermodynamic framework for dispersions of paramagnetic particles assembling in magnetic fields, taking into account mutual polarization, where the dipole moments that determine what phases self-assemble are induced by both the externally applied field and a local field arising from the dipoles induced in all of the surrounding particles. We computed the equilibrium phase diagram by theoretically constructing an expression for the free energy of the dispersion and then minimizing at thermodynamic equilibrium. We also constructed the phase diagram using Brownian dynamics simulations of mutually polarizable particles. Both the theoretical predictions and the Brownian dynamics simulations agree with the phase behavior observed in experiments. The new model predicts a eutectic line, where two crystal phases and a fluid phase simultaneously coexist. As the particles assemble, they tend to arrest in defective, percolated metastable states rather than reach their equilibrium crystal phase, especially if the field-induced forces are much larger than thermal forces. Experiments have shown that toggling the magnetic field on and off cyclically in time can avoid arrested phases and rapidly crystallize the dispersion. While the field is off, thermal motion helps anneal defective, arresting configurations that persist otherwise. Because of the cyclic driving force, the dispersion evolves until it reaches a periodic-steady-state (PSS), and the assembled structures are active and no longer at thermodynamic equilibrium. New structures, which are not necessarily stable at thermodynamic equilibrium, can be stabilized at periodic-steady-state in toggled fields. The active assembly dynamics as well as the terminal phases at PSS are controlled with parameters of the toggle protocol; however, there are no theories to describe the active assembly process in terms of toggle parameters. Using the same computational tools as with steady fields, we investigated the active assembly of mutually polarizable particles in toggled magnetic fields using

Task Progress:	Brownian dynamics simulation and theory. The assembled structures we observed as well as their assembly kinetics matched that seen in experiments. In the vicinity of the eutectic, we observed a new, non-equilibrium body-centered-orthorhombic (BCO) crystal phase. The lattice parameters of this active BCO phase respond to changes in the toggle duty. allowing one crystal to transmute readily into another. Building off our work with dispersions assembling via toggled short-ranged, isotropic attractions, we used linear irreversible thermodynamics to develop a predictive theory describing the self-assembled states in the active dispersions as functions of parameters of the toggling protocol. These predictions agree well with the phase behavior in our simulations. Another mode of active assembly involves rotating the magnetic field in time. Like the toggled assemblies, the cyclic driving force of the rotating field drives the dispersion to an active periodic-steady-state that does not need to correspond to a thermodynamic equilibrium phase. Though new crystalline phases can be assembled at large field strengths and large particle concentrations, we investigated the assembly and dynamics of dilute dispersions (one to five percent particles by volume), where the particles chain up and rotate with the external field. The mean chain length decreases with increasing rotation frequency, reflecting the balance between the magnetic torque aligning chains with the field, the hydrodynamic drag torque opposing the chains' motion, and the intrachain dipolar forces preventing particles from being sheared off. This offers a compelling strategy for magnetophoretically transporting paramagnetic particles through porous media, say through biological tissue to deliver cargo to a target site. If a magnetic field gradient is applied with the field orientation held steady in time, the magnetophoretic driving force (proportional to the strength of the induced dipole) is coupled to the interparticle forces promoting particle aggregation (proportional to the square of the induced dipole). It is desirable to have large magnetophoretic driving forces to quickly transport the particles, but the strong interparticle forces drive formation of large aggregates that have difficulty navigating (or even entering) the porous media, so the net particle flux in steady fields is quite small. If the magnetic field gradient is maintained but the field orientation is rotated in time, the magnetophoretic driving force can remain large while the rotation frequency can be tuned to suppress aggregation. With chains broken up, particles can easily navigate the small pores, and flux through porous media is enhanced. We find that the fastest particle transport occurs at intermediate rotation frequencies. Although longer chains move more slowly through pores, they can generate large mixing flows as they rotate with the field (effectively functioning as magnetic stir bars). The competition between decreasing "stir bar" length and increasing "stir speed" with increasing frequency leads to a maximum in the long-time self-diffusivity of the dispersion at intermediate rotation frequencies. This increased diffusion helps particles navigate tortuous pores, increasing the flux through porous media.
Bibliography Type:	Description: (Last Updated: 04/30/2025)
Articles in Peer-reviewed Journals	Sherman ZM, Ghosh D, Swan JW. "Field-directed self-assembly of mutually polarizable nanoparticles." Langmuir. 2018 Jun 19;34(24):7117-34. https://doi.org/10.1021/acs.langmuir.8b01135 , Jun-2018